IOAI Platform Q&A

1.
Q. VSCode does not have SSH, Jupyter, or Python extensions. / Kernel does not appear or remains in a "loading" state.

A. Problem solved. If the problem persists, Contestants now are able to install the extensions missing only on the above list. More detailed instructions can be found in the Appendix A below.

2.
Q. Nodes cannot start running (in the state of waiting or preparing) / Nodes take too long to create / Nodes disappeared unexpectedly / Nodes are automatically deleted after reconnecting.
A. Problem solved. The above problem appears due to GPU resource insufficiency yesterday, since the team-challenge groups do not release these GPU resources.

3.
Q. Nodes fail to connect to Jupyter Notebook in Bohrium.

A. Problem solved. **If the problem persists**, below are the steps to take.

nohup jupyter lab --allow-root --notebook-dir=/ --ServerApp.token=123456 --ip=0.0.0.0 --port=8888 --no-browser > jupyter.log 2>&1 &

Move to Bohrium's Node Interface. Click the "Connect" Button on the bottom right corner of the node box. Choose "Open Web Shell". Enter the following code in the Terminal shown.

4.
Q. HuggingFace failed to download models.

A. Please refer to the document **Additional Message by ISC.** Technically, we only provide access links under the huggingface.co URL. If your download triggers other domains that are not on the whitelist, you cannot access them through your laptop.

5.
Q. HuggingFace models took too long to download. / VSCode crashed while downloading models.

A. There are some complicated reasons we cannot resolve. One main technical reason is that you cannot access the website out of China. The final statement is in **Additional Message by ISC.**

6.
Q. Baseline cannot be opened properly. Refreshing results in a white screen.

A. **Contestants need to login again using English as the language used** and close the auto-translate feature in the browser.

7.
Q. The same notebook produces different scores across submissions.

A. This is a common behavior in machine learning due to non-deterministic elements like unseeded randomness in the code. Please refer to the **Additional Message by ISC** to learn how to fix it when you use Pytorch to build models.

8.
Q. Can the Contestants see their real-time ranking after submission.

A. Yes, After submitting their notebook, contestants are able to see their current ranking, score, and the current highest score.

9.
Q. Can Contestants access problem descriptions before exam begins

A. No, Opening the competition webpages will results in “This page does not exist.” With a “Back to Home” button

10.
Q. Why the Youdao translator is banned. However can I do the translation?
A. Some contestants find the bugs that Youdao can communicate with each other. So we ban them. The only way to translate during the individual contest is to use LLM embedded in Bohrium, or wait for your Team Leader’s translation.

11. If a large-scale technical problem occurred, documents containing solutions will be sent through network. Contestants will be notified to check via broadcast.

APPENDIX A

Check Extension Installations

To use SSH, please check if the following extensions are installed (Search in the Extensions sidebar in VSCode):
- Python
- Remote SSH
- Remote Explorer
- Jupyter

Once you make sure they are installed on your local instance of VSCode, please continue to the next steps.

Connect Via SSH

Use the Remote Explorer on the left sidebar to connect to the SSH host. Your host and password can be found in the connect tab in your node on bohrium.

Install Necessary Extensions on Remote Node

Now you need to install the missing extensions on the vscode server on the node. When you connect into the node, open a notebook and click "Select Kernel". It should prompt you to install the extensions. If it's not working properly, you can search in the Extensions sidebar and install Python and Jupyter from there. After installing, you should be able to select the kernel and run the notebooks.

Troubleshooting

Sometimes VSCode can get stuck displaying the "installing" part when installing extensions, press fn+f1 and type reload window to reload the VSCode window. This should fix the issue.

Additional Message by ISC

Fixing the random seed

Use the following method to fix the seed of your notebook, to ensure reproducible results across runs (you are free to choose the seed value yourself):

```
import random
import numpy as np
import torch

seed = 42

random.seed(seed)                # Python built-in random
np.random.seed(seed)              # NumPy
torch.manual_seed(seed)           # PyTorch (CPU)
torch.cuda.manual_seed(seed)      # PyTorch (single GPU)
torch.cuda.manual_seed_all(seed)  # PyTorch (all GPUs)

# Ensures deterministic behavior
torch.backends.cudnn.deterministic = True
torch.backends.cudnn.benchmark = False
```

Clarification about pre-trained models availability and usage

A separate environment is set up for each task. All environments contain the same libraries, all environments contain the 3 ResNet models listed below, and each environment optionally contains some additional cached pre-trained models. The details are below:

Task 1 environment contains:

- No additional pre-trained models

Task 2 environment contains:

- A custom pre-trained model (base.pth) provided by the task developer
- Usage examples can be found in baseline.ipynb.

Task 3 environment contains:

- A number of pre-trained models, listed in the task description
- Usage examples can be found in baseline.ipynb.

For tasks 1 and 2, you are required to submit only a notebook. Your solution to these tasks must rely on the available pre-trained models (cached or task-specific).

For task 3, you are required to submit a notebook and a model. The model you submit can be based on one of the cached pre-trained models or another model you download from HuggingFace/torchvision. Just keep in mind that:

- You are not allowed to download models over 1B parameters in size (check the Model Size tab on the right size of a model HuggingFace page)
- The model you submit as part of your solution must not exceed 2GB in size.
- If you encounter an **error message** while trying to download a model from HuggingFace, or find that a model takes too long to load, this may be an access issue that we are not able to resolve.

Details about the ResNet models, available in all task environments:

Models available:

- ResNet18
- ResNet34
- ResNet50

Usage example for ResNet18:

```
from torchvision.models import resnet18, ResNet18_Weights
model = resnet18(pretrained = ResNet18_Weights)
```

The fact that the models are available in all task environments does **not** imply that all tasks require the use of these models.

Final notes

The Bohrium environment and evaluation environments are configured to be the same. Any changes made to the Bohrium environment by contestants will not affect or be reflected in the evaluation environment.

If a task description on Bohrium is unclear or contradicts any of the statements above, assume the above is the correct information.

Since some languages are not supported by the contest laptops, the laptops will be configured to have only English (the main contest language) by default.

All contestants will be given new accounts tomorrow so they will not be able to access their files from earlier contests.

The Discussion tab in Bohrium will be disabled to prevent abuse. All clarification requests will be received through the paper form.